

Deep Dive into the KDD Process and Data Mining Techniques

- ▶ **Understanding Each Stage of Knowledge Discovery in Databases and Core Data Mining Methods**
- ▶ **Course: Data Mining and Data Warehousing**
- ▶ **Presented by: Aamir Mir**

Introduction to KDD

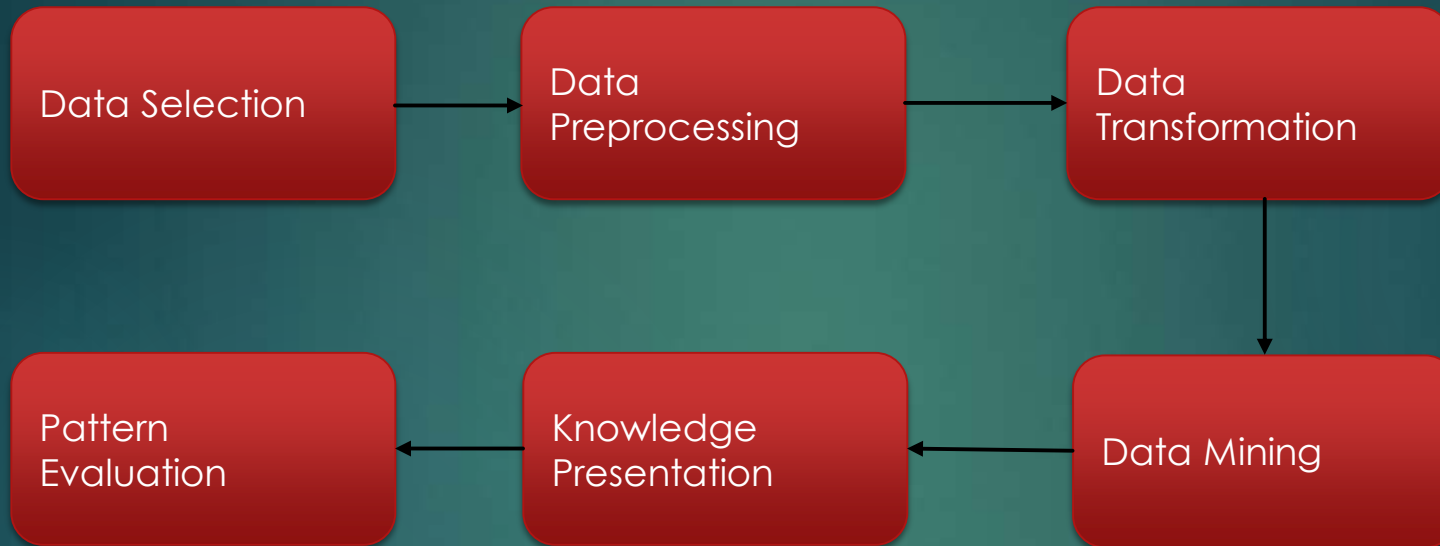
- ▶ KDD stands for Knowledge Discovery in Databases
- ▶ Complete process of discovering useful knowledge from data
- ▶ Data mining is just one step
- ▶ Goal: Extract patterns and knowledge from large data reliably

Overview of KDD Steps

- ▶ 1. Data Selection
- ▶ 2. Data Preprocessing (Cleaning)
- ▶ 3. Data Transformation
- ▶ 4. Data Mining
- ▶ 5. Pattern Evaluation
- ▶ 6. Knowledge Presentation

- ▶ Each step is crucial for success

KDD Process Flowchart



Step 1 - Data Selection

- ▶ Identify and retrieve relevant data
- ▶ Choose data aligned with objectives
- ▶ Activities: Domain study, source identification, extraction
- ▶ Challenges: Integration issues, privacy concerns

Step 2 - Data Preprocessing (Cleaning)

- ▶ Prepare clean data for analysis
- ▶ Handle missing values, remove duplicates
- ▶ Correct inconsistencies and remove noise
- ▶ Tools: OpenRefine, Pandas, Excel

Step 3 - Data Transformation

- ▶ Convert data for mining
- ▶ Techniques: Normalization, Aggregation, Encoding, Feature engineering
- ▶ Goal: Improve quality and accuracy
- ▶ Example: Age → Age groups

Step 4 - Data Mining

- ▶ Core step of KDD
- ▶ Apply algorithms to extract patterns
- ▶ Techniques: Classification, Clustering, Association, Regression, Anomaly Detection
- ▶ Choice depends on data and goals

Step 5 - Pattern Evaluation

- ▶ Identify useful and interesting patterns
- ▶ Use metrics: Accuracy, Usefulness, Novelty, Simplicity
- ▶ Tools: Confusion Matrix, ROC Curve, Lift Charts

Step 6 - Knowledge Presentation

- ▶ Visualize and communicate knowledge
- ▶ Charts, graphs, reports, dashboards
- ▶ Tools: Tableau, Power BI, Matplotlib
- ▶ Make insights understandable and actionable

Deep Dive into Data Mining Techniques

- ▶ Classification: Predict labels (SVM, Decision Trees)
- ▶ Clustering: Group similar data (K-Means, DBSCAN)
- ▶ Association: Rules like $A \rightarrow B$ (Apriori, FP-Growth)
- ▶ Regression: Predict values (Linear, Ridge)
- ▶ Anomaly Detection: Detect outliers (Isolation Forest)

Summary of KDD Process & Data Mining Techniques

- ▶ KDD is iterative and multi-step
- ▶ Data Mining is key to extracting insights
- ▶ Data quality and evaluation are vital
- ▶ Proper technique = better decisions and outcomes

Classification

- ▶ Definition: Supervised learning to predict categorical labels.
- ▶ Goal: Assign input data to predefined classes.
- ▶ Algorithms: SVM, Decision Trees, Random Forest.
- ▶ Example: Email Spam Detection, Medical Diagnosis.

Clustering

- ▶ Definition: Unsupervised learning to group similar data.
- ▶ Goal: Identify inherent structure in data.
- ▶ Algorithms: K-Means, DBSCAN.
- ▶ Example: Customer Segmentation, Market Basket Analysis.

Association

- ▶ Definition: Discover interesting relationships among variables.
- ▶ Goal: Find rules like “If A, then B”.
- ▶ Algorithms: Apriori, FP-Growth.
- ▶ Example: Market Basket Analysis, Web Usage Mining.

Regression

- ▶ Definition: Supervised learning to predict continuous values.
- ▶ Goal: Estimate numeric output from inputs.
- ▶ Algorithms: Linear Regression, Ridge Regression.
- ▶ Example: House Price Prediction, Stock Market Forecasting.

Anomaly Detection

- ▶ Definition: Identify rare or unusual patterns in data.
- ▶ Goal: Detect outliers, fraud, or faults.
- ▶ Algorithms: Isolation Forest, One-Class SVM.
- ▶ Example: Credit Card Fraud Detection, Intrusion Detection.